

■ LightSources

`LightSources` is an option to `Graphics3D` and related functions that specifies the properties of point light sources for simulated illumination.

The basic form is `LightSources -> { $s_1$ ,  $s_2$ , ...}`, where the s_i are the specifications for each light source. Each s_i has the form `{direction, color}`. The direction is specified as `{x, y, z}`, where the components are with respect to the final viewing plane. The x and y axes are in the plane; z is orthogonal to the plane; positive z is in front. Only the relative magnitude of the components is relevant; the overall normalization of the vector is ignored. The color can be specified by `GrayLevel[graylevel]` or `RGBColor[red, green, blue]`. ■ Simulated illumination determines the shading of polygons in three-dimensional pictures. ■ The shading of a particular polygon is computed as a sum of contributions from point light sources, plus a contribution from ambient light. ■ If `GrayLevel` or `RGBColor` has been specified for a particular polygon, the value will be used to give the reflectivity of the polygon. If no such specifications have been made, the polygon is taken to be a perfect (diffuse) reflector. ■ Polygons are assumed to be diffuse reflectors, and to follow Lambert's law. The intensity of reflected light is taken to be proportional to the cosine of the angle between the normal to the polygon and the direction to a particular light source. ■ See page 129. ■ See also: `AmbientLight`, `ViewPoint`.